

Sales & Customer Behavior Analytics Report

1. Executive Summary

This project analyzes sales performance and customer purchasing behavior using the Brazilian E-Commerce (Olist) dataset. The objective was to identify sales trends, customer preferences, product performance, payment behavior, and opportunities for business growth through data analytics.

The dataset was cleaned, merged, and analyzed using Python, Pandas, and data visualization techniques. Additional business metrics such as Revenue, Profit, Delivery Time, Customer Lifetime Value (CLV), and Repeat Customer Rate were calculated to provide actionable insights.

2. Project Objectives

The primary objectives of this project were:

- Analyze sales trends over time.
 - Understand customer purchasing behavior.
 - Identify top-performing product categories.
 - Evaluate payment preferences.
 - Measure customer retention and loyalty.
 - Generate business recommendations based on data.
-

3. Dataset Description

The project used the Brazilian E-Commerce (Olist) dataset consisting of:

- Customers Dataset
- Orders Dataset
- Order Items Dataset
- Order Payments Dataset
- Products Dataset

After data cleaning and merging:

- Total Records: 115,034
 - Total Features: 34
-

4. Data Cleaning and Preparation

The following preprocessing steps were performed:

Missing Value Handling

- Removed records with missing critical payment information.
- Filled missing product category information with appropriate values.
- Handled missing product dimension attributes using median values.

Date Conversion

- Converted order timestamps into datetime format.
- Calculated delivery duration in days.

Feature Engineering

Created new business metrics:

Revenue

$\text{Revenue} = \text{Price} \times \text{Quantity}$

Profit

$\text{Profit} = \text{Revenue} \times 20\%$ (assumed profit margin)

Delivery Days

$\text{Delivery Days} = \text{Delivery Date} - \text{Purchase Date}$

Customer Lifetime Value (CLV)

$\text{CLV} = \text{Total Revenue Generated by Customer}$

5. Sales Analysis

Monthly Revenue Trend

The monthly sales analysis revealed significant fluctuations throughout the year.

Key Findings

- Highest Revenue Month: May

- Revenue Generated: 1,540,926.43
- Lowest Revenue Month: September
- Revenue Generated: 644,282.48

Interpretation

Sales performance was strongest between March and August, indicating a period of increased customer demand. Revenue dropped significantly during September and October, suggesting possible seasonal effects or reduced marketing effectiveness.

Product Category Performance

Top Product Categories by Revenue:

Rank	Category	Revenue
1	Beauty & Health	1,271,413.18
2	Watches & Gifts	1,213,162.80
3	Bed, Table & Bath	1,077,968.13
4	Sports & Leisure	990,486.72
5	IT Accessories	918,837.87

Interpretation

Beauty & Health products generated the highest revenue among all categories. Consumer demand was also strong for lifestyle, home, and technology-related products.

6. Payment Analysis

Payment methods used by customers:

Payment Method	Orders
Credit Card	84,896
Boleto	22,362
Voucher	6,123
Debit Card	1,653

Key Insight

Credit Card was the dominant payment method, accounting for the majority of transactions.

Business Implication

The company should continue optimizing credit card payment options and installment-based payment services, as customers strongly prefer digital payment methods.

7. Customer Behavior Analysis

Repeat Customer Rate

Repeat Customer Rate = 15.02%

Interpretation

Approximately 15% of customers made more than one purchase, while 85% purchased only once.

Business Implication

Customer retention remains a major growth opportunity. Increasing repeat purchases could significantly improve revenue without acquiring new customers.

Customer Lifetime Value (CLV)

Customer Lifetime Value analysis identified a small group of high-value customers contributing a significant share of total revenue.

Business Implication

These customers should be targeted with:

- Loyalty programs
 - Personalized recommendations
 - Exclusive discounts
 - Premium membership offers
-

8. Delivery Performance Analysis

Delivery time was calculated using order purchase and delivery dates.

Importance

Delivery performance directly affects:

- Customer satisfaction
- Repeat purchases
- Brand reputation

Reducing delivery delays can improve customer retention and overall business performance.

9. Key Business Insights

1. May was the highest-performing sales month.
 2. September generated the lowest revenue.
 3. Beauty & Health was the most profitable product category.
 4. Credit Card was the preferred payment method.
 5. Repeat Customer Rate was 15.02%.
 6. A small number of customers generated a large share of total revenue.
 7. Customer retention presents significant growth potential.
-

10. Recommendations

Sales Strategy

- Increase inventory for high-performing categories.
- Promote top-selling products during peak seasons.

Customer Retention

- Introduce loyalty reward programs.
- Offer personalized discounts to repeat customers.
- Implement targeted email marketing campaigns.

Payment Strategy

- Improve credit card payment experiences.
- Offer installment payment promotions.

Operational Improvement

- Monitor and reduce delivery delays.
 - Improve logistics performance during high-demand months.
-

11. Conclusion

This project successfully analyzed sales and customer behavior using real-world e-commerce data. The analysis revealed important insights regarding customer preferences, revenue trends, product performance, and payment behavior.

The results show that revenue is heavily influenced by product category demand and customer purchasing patterns. By focusing on customer retention, top-performing product categories, and operational efficiency, the business can improve profitability and long-term growth.

The developed analytics dashboard further enables real-time monitoring of key business metrics and supports data-driven decision-making.